

K-Means Clustering: Numerical Walkthrough

K-Means is an iterative, partition-based clustering algorithm. It groups data points into **K** distinct clusters by minimizing the distance between points and their respective cluster centers (centroids). It runs in a loop of **Assignment** (assigning points to the nearest centroid) and **Update** (moving centroids to the average center of their assigned points) until convergence.

The Sample Dataset

Imagine we have 4 retail customers with two features: Age and Spending Score (1-100):

- **Point A:** Age = 20, Spending = 80 (Young High Spender)
- **Point B:** Age = 22, Spending = 75 (Young High Spender)
- **Point C:** Age = 60, Spending = 20 (Senior Budget Shopper)
- **Point D:** Age = 62, Spending = 22 (Senior Budget Shopper)

We want to group them into **K = 2 clusters**.

Iteration 1: Starting the Loop

Step 1: Initial Centroid Placement

We randomly choose two points to act as our starting cluster centers:

- Centroid 1 (C_1): Placed at Point A (**20, 80**)
- Centroid 2 (C_2): Placed at Point C (**60, 20**)

Step 2: Distance & Cluster Assignment

We calculate the straight-line (Euclidean) distance from every point to both centroids:
$$\text{Distance} = \sqrt{(X_{\text{point}} - X_{\text{centroid}})^2 + (Y_{\text{point}} - Y_{\text{centroid}})^2}$$

- **Point A (20, 80):** Dist to \$C_1\$ is 0. Dist to \$C_2\$ is 72.1. **Verdict: Cluster 1.**
- **Point B (22, 75):** Dist to \$C_1\$ is $\sqrt{(22-20)^2 + (75-80)^2} = \sqrt{29} \approx 5.39$. Dist to \$C_2\$ is 66.85. **Verdict: Cluster 1.**
- **Point C (60, 20):** Dist to \$C_1\$ is 72.1. Dist to \$C_2\$ is 0. **Verdict: Cluster 2.**
- **Point D (62, 22):** Dist to \$C_1\$ is 71.61. Dist to \$C_2\$ is $\sqrt{(62-60)^2 + (22-20)^2} = \sqrt{8} \approx 2.83$. **Verdict: Cluster 2.**

Step 3: Recalculate Centroid Positions (Centroid Update)

Centroids shift to the exact average center of their assigned members:

- **New Centroid 1 (\$C_1\$)** (Average of A and B): Age = $\frac{20+22}{2} = 21$, Spending = $\frac{80+75}{2} = 77.5$. Coordinates: **(21, 77.5)**.
- **New Centroid 2 (\$C_2\$)** (Average of C and D): Age = $\frac{60+62}{2} = 61$, Spending = $\frac{20+22}{2} = 21$. Coordinates: **(61, 21)**.

Iteration 2: Verifying Convergence

Step 4: Recalculate Distances & Assignments

We check if any points change their cluster labels when measured against the shifted centroids \$C_1\$ (21, 77.5) and \$C_2\$ (61, 21):

- **Point A (20, 80):** Dist to \$C_1\$ is $\sqrt{(20-21)^2 + (80-77.5)^2} = 2.69$. Dist to \$C_2\$ is 71.85. Remains in **Cluster 1**.
- **Point B (22, 75):** Dist to \$C_1\$ is 2.69. Dist to \$C_2\$ is 67.90. Remains in **Cluster 1**.
- **Point C (60, 20):** Dist to \$C_1\$ is 69.90. Dist to \$C_2\$ is $\sqrt{(60-61)^2 + (20-21)^2} = 1.41$. Remains in **Cluster 2**.
- **Point D (62, 22):** Dist to \$C_1\$ is 69.11. Dist to \$C_2\$ is 1.41. Remains in **Cluster 2**.

Since no points changed their cluster assignments, the algorithm has officially **converged**!

Customer	Age, Spend	Final Cluster	Demographic Segment
Point A	(20, 80)	Cluster 1	Young High Spenders
Point B	(22, 75)	Cluster 1	Young High Spenders

Customer	Age, Spend	Final Cluster	Demographic Segment
Point C	(60, 20)	Cluster 2	Senior Budget Shoppers
Point D	(62, 22)	Cluster 2	Senior Budget Shoppers